

BARC0130: Advanced Mathematical Modelling & Analysis

Coursework 1

Topics Covered: Series & Approximation + Statistics

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QUESTION 1

Part A: The source data

The data chosen to be the most suitable candidate going forward was derived from the signals measured from mycelium samples stimulated at 100 Hz, 1 kHz and 10kHz. The distinct pattern of the 100 Hz reading (figure 1) stood out for its unique and repeatable piecewise continuous form and what it represented as a measure of the electrical response in fungal structures. [1] Living mycelial networks are known to have properties of memristors, capacitors and other sensors and can reliably transfer signals. Therefore, further research into their properties may be useful. Hopefully, the analysis of this data will provide some insight into the properties and functions of these mycelial networks.

Figure 1. Representative output signals (left), recovered amplitudes from output signals (centre) and cross spectral densities from output signals (right), from three separate sets of mycelia stimulated with 100 Hz, 1 KHz and 10 KHz square waves.[1]

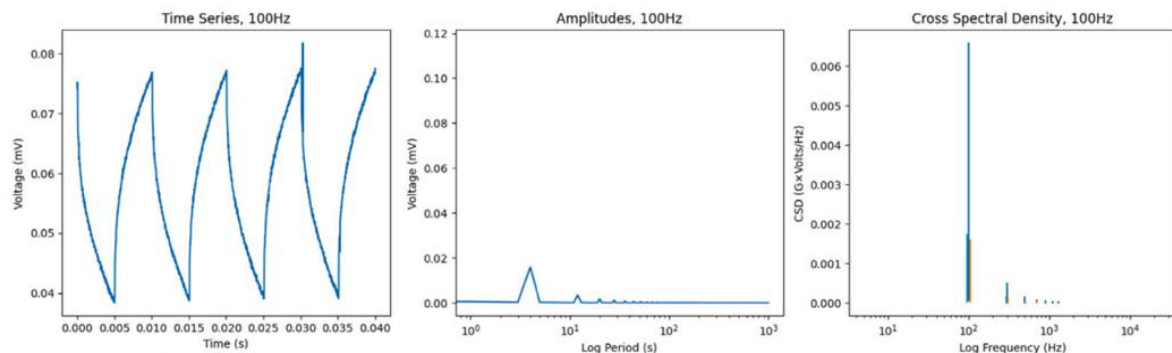
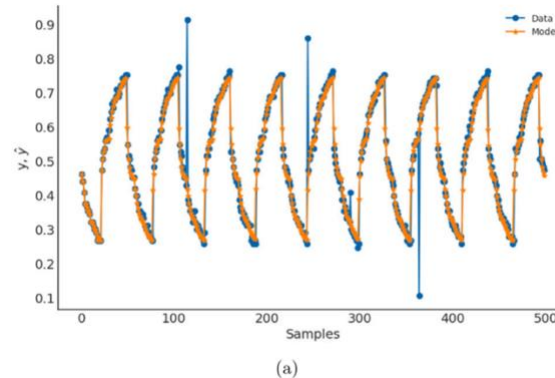
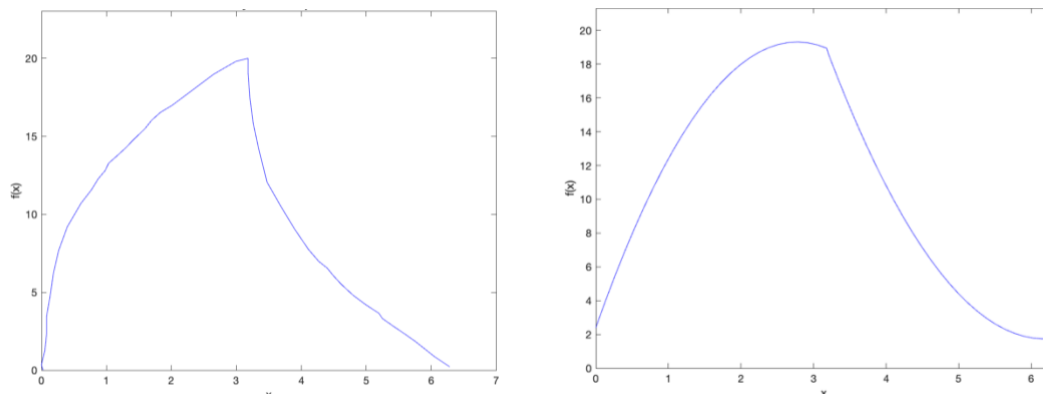


Figure 2. Representative Nonlinear AutoRegressive Exogenous (NARX) modelling of input–output transfer function, showing time series data and model predictions. (a) Test-data free run simulation, RRSE 0.255 [1]



Part B: Digitisation with Part C: Function Fitting

Figures 3 & 4. Graph of initial datapoints on the left and a graph of fitted datapoints on the right



The code below shows how the geometric data was fitted and exported as an excel file 'Mycelium_Impulse_Data1.xlsx'. The ginput function was used with an imported png of the mycelium pattern which was then fixed and sorted to represent the data more accurately. The early version of this data can be seen in figure 3 which is noticeably imprecise and unfiltered. Variables X and Y at the end of the code below are the final smoothly fitted points that were used to fit two polynomials to the graph represented in figure 4. More of this code can be seen in the attached MATLAB file but note that some of the datapoint-to-curve fitting for Task 1C was done in Task 1B (specifically in a section of the MATLAB code called 'Task 1B & Task 1C: Fitting of Polynomial Datapoints to Functions:').

```
% Capture 50 data points using ginput function
```

```
[x_points, y_points] = ginput(50);
```

```
% Curve one fitting
```

```
X_c1 = linspace(min(xpts_curve1), max(xpts_curve1), pts/2);
```

```
Y_c1 = polyval(poly_curve1_round, X_c1);
```

```
% Curve two fitting
```

```
X_c2 = linspace(min(xpts_curve2), 2*pi, pts/2);
```

```
Y_c2 = polyval(poly_curve2_round, X_c2);
```

```
% Reform data into datapoints
```

```
X = [X_c1 X_c2];
```

```
Y = [Y_c1 Y_c2];
```

```
% Converting Variable to a text file/Excel spreadsheet
```

```
T = table(X', Y', 'VariableNames', {'X', 'Y'});
```

```
writetable(T, 'Mycelium_Impulse_Data1.xlsx');
```

Part C: Modelling

In equation 1 below, the Fourier series for the electrical signals of mycelium-bound composites was determined by modelling the 100 Hz impulse data as a piecewise continuous function of $f(x) = -\frac{11}{5}x^2 + \frac{61}{5}x + \frac{12}{5}$ from zero to π and $f(x) = \frac{19}{10}x^2 - \frac{47}{2}x + \frac{372}{5}$ from π to 2π . The Fourier coefficients and the results of the final series were calculated in MATLAB (Task 1C) as well, and used to validate the accuracy of the Fourier calculations below.

The Fourier Series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\omega x + b_n \sin n\omega x) \quad n \in \mathbb{N} \text{ if harmonious}$$

$$\therefore f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \text{as } \omega = \frac{2\pi}{T}, T = 2\pi$$

Defining the Function

$$f(x) = -\frac{11}{5}x^2 + \frac{61}{5}x + \frac{12}{5} \quad 0 \leq x < \pi$$

$$f(x) = \frac{19}{10}x^2 - \frac{47}{2}x + \frac{372}{5} \quad \pi \leq x < 2\pi \quad (1)$$

$$f(x) = f(x + 2\pi n), \text{ where } n \text{ is a whole number} = 0, 1, 2 \dots$$

The Fourier Coefficients

$$a_0 = \frac{2}{T} \int_0^{2\pi} f(x) dx$$

$$\begin{aligned} \therefore a_o &= \frac{1}{\pi} \int_0^{\pi} f(x) dx + \frac{1}{\pi} \int_{\pi}^{2\pi} f(x) dx \\ \therefore a_o &= \frac{1}{\pi} \int_0^{\pi} -\frac{11}{5}x^2 + \frac{61}{5}x + \frac{12}{5} dx + \frac{1}{\pi} \int_{\pi}^{2\pi} \frac{19}{10}x^2 - \frac{47}{2}x + \frac{372}{5} dx \end{aligned}$$

$$a_o = \frac{74\pi^2 - 583\pi + 1536}{20} = 21.74 \text{ 2 d.p.}$$

$$\begin{aligned} a_n &= \frac{2}{T} \int_0^{2\pi} f(x) \cos nx dx \\ \therefore a_n &= \frac{1}{\pi} \int_0^{\pi} f(x) \cos nx dx + \frac{1}{\pi} \int_{\pi}^{2\pi} f(x) \cos nx dx \\ \therefore a_n &= \frac{1}{\pi} \int_0^{\pi} -\frac{11}{5}x^2 + \frac{61}{5}x + \frac{12}{5} \cos nx dx + \frac{1}{\pi} \int_{\pi}^{2\pi} \frac{19}{10}x^2 - \frac{47}{2}x + \frac{372}{5} \cos nx dx \\ a_n &= \frac{76\pi - 82\pi(-1)^n + 357(-1)^n - 357}{10\pi n^2} \end{aligned}$$

$$\begin{aligned} b_n &= \frac{2}{T} \int_0^{2\pi} f(x) \sin nx dx \\ \therefore b_n &= \frac{1}{\pi} \int_0^{\pi} f(x) \sin nx dx + \frac{1}{\pi} \int_{\pi}^{2\pi} f(x) \sin nx dx \\ \therefore b_n &= \frac{1}{\pi} \int_0^{\pi} -\frac{11}{5}x^2 + \frac{61}{5}x + \frac{12}{5} \sin nx dx + \frac{1}{\pi} \int_{\pi}^{2\pi} \frac{19}{10}x^2 - \frac{47}{2}x + \frac{372}{5} \sin nx dx \end{aligned}$$

$$\begin{aligned} A_n &= -357\pi n^2 + 41\pi^2 n^2 + 720n^2 - 82 \\ B_n &= -76\pi^2 n^2 - 720n^2 + 470\pi n^2 + 82 \end{aligned}$$

$$b_n = \frac{A_n(-1)^n + B_n}{10\pi n^3}$$

Final Fourier Series Values

$$\therefore f(x) = \frac{74\pi^2 - 583\pi + 1536}{40} + \sum_{n=1}^{\infty} \left(\frac{76\pi - 82\pi(-1)^n + 357(-1)^n - 357}{10\pi n^2} \cos nx + b_n = \frac{A_n(-1)^n + B_n}{10\pi n^3} \sin nx \right)$$

$$f(x) \approx \frac{74\pi^2 - 583\pi + 1536}{40} + \sum_{n=1}^5 \left(\frac{76\pi - 82\pi(-1)^n + 357(-1)^n - 357}{10\pi n^2} \cos nx + \frac{A_n(-1)^n + B_n}{10\pi n^3} \sin nx \right)$$

$$f(x) \approx 10.87 - (6.93\cos x + 0.15\cos 2x + 0.77\cos 3x + 0.04\cos 4x + 0.28\cos 5x \dots) + (5.33\sin x + 0.15\sin 2x + 0.23\sin 3x + 0.08\sin 4x + 0.06\sin 5x \dots) \text{ 2 d.p. Evaluated for 5 values of the series}$$

Where $f(x)$, a_o , a_n , a_1 , a_2 , b_n , b_1 , b_2 , n , m , T and ω are the Fourier series, the coefficient a_o , the coefficient a_n , the coefficient a_1 , the coefficient a_2 , the coefficient b_n , the coefficient b_1 , the coefficient b_2 , the n^{th} value in the series, the maximum number of values considered, the time period and the angular speed.

Figure 5. Fourier Series Approximation of the 5th 8th and 20th terms

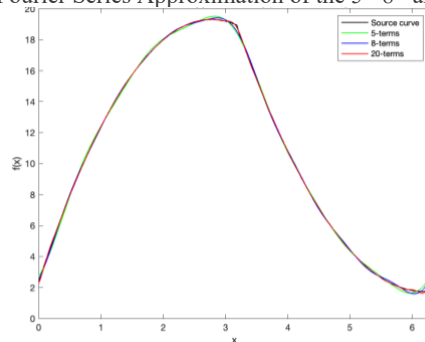


Figure 5 is a comparison of the Fourier series values at 5, 8 and 20 terms, where the 8-term series was determined as having the best fit in relation to the source data (in black on the figure above). The 8-

term series was also chosen for being sharper at the peak than the 5-term curve but also more consistent than the 20-term curve, which wavers as it nears 2π . Overall however, the Fourier series is almost indistinguishable from the source data, so much so that the differences between each variation are slight and hard to parse out. See the MATLAB code for Task 1C, and the sample code below, for more on how the graph was plotted and constructed.

Part D: Curve generation

The function below was used to calculate an 8-term Fourier series by summing all of the Fourier values for each coefficient and applying them to the Fourier series formula.

```
% Calculating Fourier series terms
function y = fourier_series(x, a0, an, bn)
    N = length(an);
    y = a0/2 * ones(size(x));
    for n = 1:N
        y = y + an(n)*cos(n*x) + bn(n)*sin(n*x);
    end
end
```

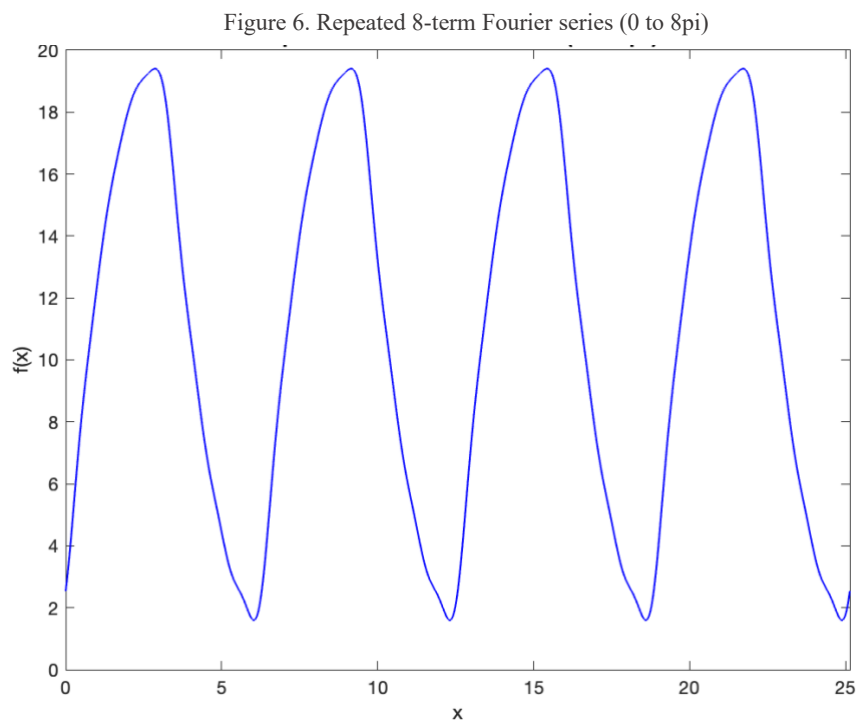


Figure 6 above contains the Fourier approximation to the mycelial pattern in the range 0 to π , and was modelled in MATLAB (Task 1D). Apart from slight irregularities near the turning points of the graph, this fit is remarkably accurate to the signal propagation patterns seen in figures 1 and 2.

QUESTION 2

Part A

The data below was cleaned in task 2A of the attached MATLAB file. The code goes into greater detail on how the data was cleaned within 3 standard deviations of the mean and rounded to the nearest millimetre. Overall, this process found two outliers that may have been the result of solar flares and removed them such that only the 148 cleaned datapoints were considered.

Table 1. The means, standard deviations and standard error for the GNSS data

GNSS Data	Equation	Value 3 d.p. (nearest mm)
Sample Mean	$\bar{X} = E(\bar{X}) = \frac{1}{n} \sum_{i=1}^n E(x_i)$	$\bar{X}_1 = 999.986$
Sample Standard Deviation	$s = s_1 = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$	$s_1 = 0.300$
Sample Standard Error	$SE(\bar{X}) = SE_1 = \sqrt{\sigma^2} = \frac{\sigma}{\sqrt{n}}$	$SE_1 = 0.025$

Where $\bar{X}$, $\bar{X}_1$, $E(\bar{X})$, n , x_i , s , s_1 and SE_1 are the sample mean, the sample mean of the GNSS dataset, the expectation, the sample size, the i^{th} value of the sample, the sample standard deviation, the sample standard deviation of the GNSS data, the population mean of the GNSS data and the standard error for the GNSS data respectively.

Part B

Below is the step by step process (equation 2) of the χ^2 goodness of fit test for the GNSS data, where the expected values were calculated in MATLAB (Task 2B). The χ^2 value was less than the critical value, meaning the normal distribution was a good fit for the GNSS data. This, along with the relatively (in relation to the geometric data) low GNSS standard deviation and standard error may suggest that the GNSS sample mean is a more accurate representation of the tower's height. Since the mean is 999.986 3 d.p, this may support concerns for the effects of vertical shortening and shrinkage on the tower. [2]

Table 2. Chi-squared critical value table [4][2]

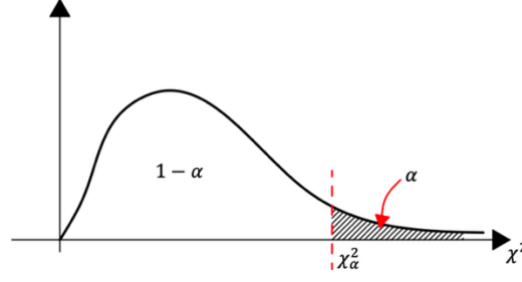
Table G	The Chi-Square Distribution										
Degrees of freedom	α										
	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005	
1	—	—	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879	
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597	
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838	

Table 3. Chi-squared expected and observed class data

Intervals (m) 3 d.p.	$999.091 \leq x < 999.497$	$999.497 \leq x < 999.904$	$999.904 \leq x < 1000.310$	$1000.310 \leq x < 1000.717$	$1000.717 \leq x < 1001.123$
Expected Freq (E_i) 3 d.p.	7.446	50.407	69.262	19.577	1.086
Observed Freq (O_i)	8	41	83	15	1

[2]

Figure 7. Visualisation of a χ^2 goodness of fit test, which is always right-handed. [2]



H_0 : the normal distribution is a good fit for the data

H_1 : the normal distribution is not a good fit for the data

$$DoF = k - 1 - m = 2 \quad \text{where } k = 5, m = 2$$

$$\chi^2_{\alpha} = (\text{From chi-square table above [4]}) = 5.99$$

$$E_i = E(\bar{X}) = \sum_{i=1}^n (x_i)P(x_i)$$

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

$$\chi^2 = \frac{(8-7.446)^2}{7.446} + \frac{(41-50.407)^2}{50.407} + \frac{(83-69.262)^2}{69.262} + \frac{(15-19.577)^2}{19.577} + \frac{(1-1.086)^2}{1.086} = 5.599 \text{ 3 d.p.} \quad (2)$$

$$|\chi^2| < \chi^2_{\alpha}$$

Therefore there isn't enough evidence to reject the null hypothesis that the normal distribution is a good fit for the data

Where DoF, k , m , E_i , O_i , n , x_i , χ^2 , χ^2_{α} are the degree of freedom, the number of classes, the number of assumed parameters (in the calculation of the expected value), the expected frequency for interval i , the observed frequency for interval i , the sample size, the i^{th} value of the sample, the chi squared statistic and the chi squared critical value, respectively.[2][4][2]

Part C

The geometric data below is cleaned and rounded to the nearest millimetre in MATLAB. The values below are to 3d.p. to reflect a precision of 1mm, the same resolution as the source data. A comparison to the GNSS data reveals that the geometric method ($h = D \tan \theta$) has a higher standard deviation and standard error. This suggests that the geometric method is a less precise and less accurate, therefore, if the GNSS receiver keeps experiencing issues (like breaking down or encountering solar flares) a new measurement method may need to be found.

Table 4. The means, standard deviations and standard error for the geometric data

Geometric Data	Equation	Value
Sample Mean	$\bar{X} = E(\bar{X}) = \frac{1}{n} \sum_{i=1}^n E(x_i)$	$\bar{X}_2 = 1001.939$
Sample Standard Deviation	$s = s_2 = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$	$s_2 = 1.018$
Sample Standard Error	$SE(\bar{X}) = SE_2 = \sqrt{\sigma^2} = \frac{\sigma}{\sqrt{n}}$	$SE_2 = 0.183$

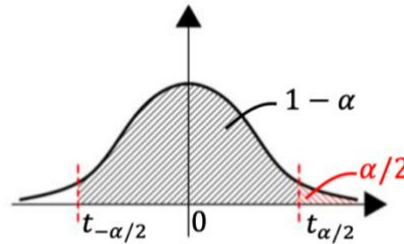
Where h , D , θ , $\bar{X}$, $\bar{X}_2$, $E(\bar{X})$, n , x_i , s , s_2 , μ_2 and SE_2 are the building height (m), the distance from the base of the tower to the position of the angle measurement (m), the angle measurement ($^{\circ}$), the sample mean, the sample mean of the geometric dataset, the expectation, the sample size, the i^{th} value of the

sample, the sample standard deviation, the sample standard deviation of the geometric data, the population mean of the geometric data and the standard error for the geometric data respectively.

Part D

The t-test below operates with a significance level (α) of 0.05 and assumes the data fits to a normal distribution. The t-test uses the equations in (3) below and compares them to the critical value [5] to test if the two means are significantly different from one another. Since the null hypothesis was rejected and the t-test statistic (t) was large and negative, it seems as though the GNSS mean was significantly lower than the geometric mean. [6] This difference may be a result of the geometric data's slightly lower accuracy.

[5]Figure 8. Two-tailed t-test distribution



Hypothesis t-test for the means:

H_0 : The means of the two data sets (GNSS and geometric measurements) are not significantly different from one another

H_1 : The means of the two data sets (GNSS and geometric measurements) are significantly different from one another

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

$$\text{Two-tailed test } \therefore \frac{\alpha}{2} = 0.025$$

$$t_{\alpha/2} = (\text{From statistics table[5]}) = 2.042$$

$$\text{DoF} = n_2 - 1 = 30$$

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{(999.986 - 1001.939) - (0)}{\sqrt{\frac{0.300^2}{148} + \frac{1.018^2}{31}}} = -10.586 \text{ 3 d.p.} \quad (3)$$

$$|t| > t_{\alpha/2}$$

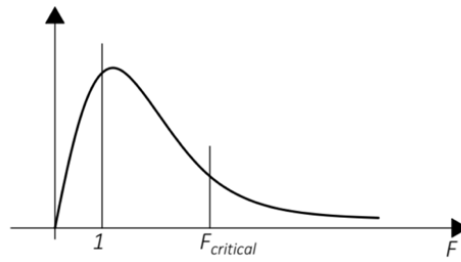
Therefore there is sufficient evidence at the 95% confidence level to reject the null hypothesis. Therefore, the means of the two data sets are significantly different from one another

Where H_0 , H_1 , μ_1 , μ_2 , α , n_1 , n_2 , t and $t_{\alpha/2}$ are the null hypothesis, the alternative hypothesis, the mean of the GNSS data, the mean of the geometric data, the significance level, the size of the GNSS data sample, the size of the geometric data sample, the t-test statistic and the t-test critical value, respectively.

The F-test determines how different the values of the GNSS and geometric variances are and therefore if one dataset is significantly noisier than the other. Given that the null hypothesis was rejected and the variance of the geometric data was the larger of the two, the geometric data is significantly noisier than

the GNSS dataset. This, along with the higher geometric standard error only adds to the dubious quality of the geometric results.

Figure 9. F-test distribution



Hypothesis F-test for the variances:

H_0 : The variances of the two data sets (GNSS and geometric measurements) are not significantly different from one another

H_1 : The variances of the two data sets (GNSS and geometric measurements) are significantly different from one another

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$H_1: \sigma_1^2 \neq \sigma_2^2$$

$$\text{Two-tailed test } \therefore \frac{\alpha}{2} = 0.025$$

$$F_{\alpha/2} = (\text{From statistics table}[5]) = 1.55$$

(4)

$$DoF_{\text{numerator}} = n_2 - 1 = 30, \quad DoF_{\text{denominator}} = n_1 - 1 = 147 \text{ (taken as 120)}$$

$$F = \frac{s_2^2}{s_1^2} = \frac{1.018^2}{0.300^2} = 11.515 \text{ 3 d.p.} \quad \text{where } s_2^2 > s_1^2$$

$$|F| > F_{\alpha/2}$$

Therefore there is/is not sufficient evidence at the 0.05 significance level to reject the null hypothesis that the means of the two data sets are not significantly different from one another

Where $\sigma_1^2, \sigma_2^2, s_1^2, s_2^2, DoF_{\text{numerator}}, DoF_{\text{denominator}}, F$ and $F_{\alpha/2}$ are the variance of the GNSS data, the variance of the geometric data, the sample variance of the GNSS data, the sample variance of the geometric data, the degree of freedom of the geometric data, the degree of freedom of the GNSS data, the F-test statistic and the F-test critical value, respectively.

Part E

The t and F tests assume the data is independent and normally distributed. However, though the GNSS data underwent the χ^2 test and was deemed a good fit for the normal distribution, the same wasn't done for the geometric data, which was assumed to be fitting. In reality, the Jeddah tower [3] and its data (both GNSS and geometric) may be heavily impacted by unpredictable and varying environmental conditions that don't fit to a normal hypothesis test analysis. Additionally, due to concrete creep and foundation settlement the data may include long-term trends that would be better captured by temporally-dependant models like a time-series.[7][8] These types of models may also be able to differentiate between random GNSS fluctuations and structural and geological settlement patterns [9]. Overall, while acceptable, the inference techniques used in tasks A to D may be more useful for confirming the true height of the tower if they were able to model settlement trends and anomalies based on patterns in past data (like a time series analysis model can) [10].

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